

A Network and Its Reflection: What True, False, and Noise Self-Model Broadcasts Do and Do Not Change in an Evolving Multi-Agent Network

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Abstract. Many networks respond to measurements of their own state: markets to indices, organizations to metrics, platforms to trending signals. We present a controlled laboratory for this reflexive loop: an evolving agent population forms an adaptive network whose macrostate is compressed into a self-model and broadcast back — accurately, distorted, or as matched-range noise. Across 1,527 deterministic runs with paired-seed comparisons: measurement without broadcast is causally inert. A natural self-fulfilling-prophecy statistic registers strong apparent attraction toward false and replayed self-models — but a passive counterfactual, scoring untreated twins against matched references, fully accounts for it: a caution for empirical claims of performativity. What broadcasts demonstrably change is behavior and structure, and which lies come true is set jointly by the lie and the response regime. Finally, heritable attention to the broadcast declines across generations, most strongly under high-drive false and noisy feedback: populations evolve to ignore consequential misleading observers.

Keywords: adaptive networks, agent-based modeling, reflexivity, self-fulfilling dynamics, performative prediction, evolution of trust

1 Introduction

Adaptive networks — graphs whose topology coevolves with the dynamical states of their nodes — are a central object of network science [1, 2]. A distinctive subclass has received far less controlled attention: networks that respond to *descriptions of themselves*. Financial markets move on published indices computed from their own trades [3]; universities restructure around rankings derived from their own behavior [4]; recommendation platforms amplify items *because* an internal measurement declared them trending. In each case a macroscopic measurement is compressed, published, and re-enters as an input to microscopic decisions — the loop Merton called the self-fulfilling prophecy [5], whose measure-distorting variant is Goodhart’s law [6].

Empirical study of this loop is confounded: one can rarely withhold the measurement, falsify it in controlled directions, or replay history under identical randomness. We therefore built a minimal simulated world where the loop

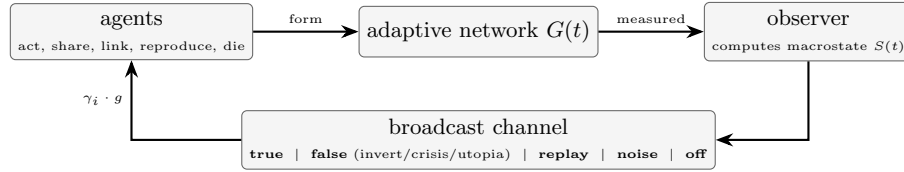


Fig. 1. The reflexive loop and its corruption points. Local actions generate the network; the observer compresses it into a self-model; the broadcast channel returns it truthfully, distorted, replayed from another run, as noise, or not at all. Action-propensity responses are scaled by heritable sensitivity γ_i and global gain g ; the hub-targeted pruning probability is scaled by γ_i alone.

can be opened, closed, and corrupted at will, every run is exactly reproducible from (configuration, seed), and agents are transparent trait vectors rather than learned policies, so every pathway from broadcast to response is inspectable.

Our contributions are: (1) an open, deterministic experimental platform coupling an evolving agent ecology, an adaptive interaction network, a global observer, and a controllable broadcast channel; (2) a six-condition causal design — verifying observer-side-effect invariance and isolating the effect of broadcasting, with matched-range noise and replayed-self-model controls — plus a *passive-counterfactual methodology* for reflexivity metrics: a natural “story pull” statistic registers strong apparent steering that untreated twin trajectories fully account for, a direct caution for empirical tests of performative prediction; (3) evidence for what broadcasts *do* change: truthful feedback doubles costly cooperation, distorted broadcasts mobilize or pacify depending on the lie–response pairing (reversing under conformist responses), and broadcasts densify the network while false and noise feedback reduce hub concentration; and (4) receiver sensitivity modeled as a continuous heritable trait, whose *population distribution shifts systematically under feedback*, tracking the broadcast-implied corrective-drive intensity and surviving two architectural controls — trust in the observer is not updated by any rule, it evolves. Figure 1 summarizes the design.

2 Related Work

Adaptive and temporal networks. Coevolution of state and topology [1, 2] covers our setting (births, deaths, rewiring by agent decisions); robustness to targeted node removal is classical [7], and we use hub removal as a standardized perturbation. **Global coupling.** A mean field fed back to units reshapes collective dynamics [8]; here that field is multidimensional, potentially *false*, and units are born, die, and evolve. **Reflexivity and reactive measurement.** Self-fulfilling prophecy [5], reflexivity in markets [3], and the reactivity of rankings and metrics [4, 6] motivate our false-broadcast conditions; here they are dosable and ablatable. **Performative prediction and endogenous signals.** Predictions that alter the distribution they forecast are formalized as performative

prediction [19, 23], including networked settings [20] and populations deciding whether to trust predictions of collective outcomes [21] — there trust is *updated* from past accuracy, whereas here attention is inherited and mutated with no rule referencing reliability: trust is not updated, it evolves; self-fulfilling signals of an endogenous network state arise in congestion games [24]. Causal identification of performative effects without randomization is an active problem [22]; ours is narrower and complementary — a concrete false-positive failure mode of directional-alignment metrics, diagnosed with paired passive twins (Sect. 5.2), plus turnover and a heritable receiver-sensitivity trait absent from these models. **Signaling and the evolution of trust.** In sender–receiver games receivers can evolve to ignore unreliable signals [9]; here the signal is *self-referential* and attention to it is a continuous heritable trait in an energy economy, not a payoff matrix. **Agent-based modeling.** Collective opinion and cooperation are mainstays [16, 17]; we follow the generative ABM tradition [10, 11], with an engine using ideas from Mesa [12] and NetworkX [13]. Multi-agent RL treats global state as an *architectural* aid [14]; we treat it as an *experimental variable* with controlled veracity. Language-model agent populations increasingly study emergent social behavior [15]; this rule-based setting provides causally clean baselines for the shared-dashboard designs natural there.

3 The Model

3.1 Agents and ecology

Each agent i carries a heritable trait vector $\theta_i = (\text{coop}, \text{expl}, \text{risk}, \text{soc}, \text{share}, \gamma_i) \in [0, 1]^6$, where γ_i is its *global sensitivity* — how strongly broadcasts modulate its behavior. Agents hold energy; each step they pay metabolism and choose one action — harvest, share (costly transfer to a neighbor), connect, prune, reproduce, or idle — with probabilities proportional to trait- and state-dependent propensities. The resource pool regenerates logistically; reproduction requires an energy threshold and copies traits with Gaussian mutation ($\sigma = 0.05$); energy exhaustion or age causes death. Founding ages are staggered to avoid cohort artifacts.

3.2 Observer and broadcast conditions

Every $\Delta t = 10$ steps the observer computes the macrostate $S(t)$: population, fragmentation (1 minus largest-component share), degree- and betweenness-concentration, Freeman centralization [18], per-capita costly-helping rate, energy Gini, and turnover, plus mean traits (logged, never broadcast). The broadcast $b(t)$ carries five of these, normalized to $[0, 1]$: *fragmentation*, *degree concentration* (top-5% degree share; distinct from Freeman centralization), *cooperation*, *inequality*, and *turnover*. Six conditions define what agents receive (Table 2). Distorted broadcasts come in three modes: *invert* ($b_k = 1 - s_k$ componentwise), *crisis* (fixed alarming values: fragmentation 0.9, degree concentration 0.9, cooperation 0.1, inequality 0.9, turnover 0.9), and *utopia* (reassuring: 0.05, 0.2, 0.9, 0.1, 0.2).

Table 1. Model parameters (defaults; campaign overrides noted in the text).

initial population	150	observer interval Δt	10 steps
population cap	1200	feedback gain g	0.8 (swept 0.2–1.6)
initial edges per agent	3 (random)	mutation σ	0.05
initial energy	20	initial traits	mean 0.5, s.d. 0.15
resource capacity K	6000	reproduction threshold / cost	30 / 16
base regeneration	60/step	min. reproduction age	15 steps
regeneration rate r	0.12	lifespan (mean $\pm$ s.d.)	400 ± 80 steps
harvest cap	2.5/step	connect cost / share amount	0.25 / 3.0
metabolism	0.8/step	run length	3000–4000 steps
shock time t_{shock}	2000 (1500 in the $2\times$ scale check)		

Table 2. Experimental conditions. All start from matched parameter distributions with paired random seeds.

Condition	Observer	Agents receive	Role
A — Local only	off	nothing	baseline
B — Observed, blind	on	nothing	measurement control
C — True feedback	on	accurate $S(t)$	main treatment
F — False feedback	on	distorted $S(t)$	reflexivity probe
N — Noise feedback	on	uniform random, matched ranges	stimulation control
R — Replayed self-model	on	another run’s genuine $S(t)$	structure control

3.3 Response model

Each step, agent i selects one action a with probability proportional to

$$w_i(a, t) = w_a(\theta_i, x_i(t)) + \gamma_i g f_a(b(t)), \quad (1)$$

where $w_a(\theta_i, x_i)$ is the trait- and local-state-dependent base propensity, g is the global feedback gain, and f_a is the response rule. In the primary *corrective* regime agents repair reported deficits: $f_{\text{connect}} = \max(0, b_{\text{frag}} - 0.3)$, $f_{\text{share}} = \max(0, 0.5 - b_{\text{coop}}) + 0.5 \max(0, b_{\text{ineq}} - 0.5)$, $f_{\text{harvest}} = \max(0, b_{\text{turn}} - 0.5)$, and $f_{\text{prune}} = 0.3 \max(0, b_{\text{cent}} - 0.6)$. In the *conformist* regime agents imitate the reported norm instead: $f_{\text{share}} = b_{\text{coop}}$, $f_{\text{prune}} = 0.5 b_{\text{frag}}$, $f_{\text{connect}} = \max(0, 0.6 - b_{\text{frag}})$, and $f_{\text{harvest}} = b_{\text{ineq}}$. Two further mechanisms act on *target selection* rather than action choice, in both regimes: while reported degree concentration exceeds 0.6, a pruning agent cuts its highest-degree neighbor (ties by id) with probability γ_i (g scales action propensities only, not this probability), else a uniform neighbor; and new links attach to a neighbor-of-a-neighbor with probability 0.7 when available (else a uniform non-neighbor), an explicit triadic-closure tendency behind the clustering in Sect. 3.4. All rule constants are fixed across all experiments; Table 1 lists the model parameters.

3.4 Emergent network structure

The emergent networks are dense, clustered, and heterogeneous: at $t = 2,000$ the giant component has $L \approx 1.8\text{--}2.5$, clustering $1.8\text{--}2.9\times$ a matched Erdős–Rényi null, and degree coefficient of variation $0.6\text{--}0.74$; against the stricter degree-preserving (configuration-model) null most of the clustering excess traces to the degree sequence (ratios $1.0\text{--}1.6$), so we make no small-world claim. Feedback acts on a small, dense, degree-heterogeneous clustered graph, not a homogeneous mixing population.

3.5 Reproducibility

The simulation is deterministic given (configuration, seed): one PRNG drives all behavioral choices and a separate dedicated stream generates stochastic broadcast signals, so constructing the noise signal never advances the behavioral PRNG; agents act in sorted order and a state hash certifies replay (identical hashes verified across machines and operating systems). All code, configurations, and analysis scripts are public.

4 Experimental Design

The standardized shock removes, in a single step at $t_{\text{shock}} = 2000$ (except in the $2\times$ scale check: $t_{\text{shock}} = 1500$ in a 3,000-step run), the highest-degree living agents (ties broken by agent id): a fixed 10 hubs in the mild variant, the top $\max(1, \lfloor 0.4 |V_{\text{live}}| \rfloor)$ in the harsh variant. Four outcome *families* were fixed before the first campaign: (1) post-shock recovery time — the first $t > t_{\text{shock}}$ at which the living population regains 90% of its pre-shock baseline, defined as its mean over $[t_{\text{shock}} - 500, t_{\text{shock}})$; (2) mean post-shock fragmentation; (3) per-capita costly-helping rate, averaged after a 500-step burn-in; (4) self-model correspondence, operationalized as *story pull*: with $s_k(t)$ the value of macrostate component k at observer tick t , $b_k(t)$ the broadcast value, and $\varepsilon = 0.02$,

$$P = \left\langle \text{sign}(b_k(t) - s_k(t)) (s_k(t+1) - s_k(t)) \right\rangle_{(t,k): |b_k(t) - s_k(t)| > \varepsilon}, \quad (2)$$

averaged over all qualifying (tick, component) pairs across the five broadcast components; $P > 0$ means the macrostate moves toward the reference where the two differ. The threshold ε makes P well defined under false and noise broadcasts (and undefined for C, whose broadcast equals the current state).

Outcome pre-specification. Eq. 2 is a *post hoc* directional refinement of the fourth family, adopted because the original distance-based form is degenerate under truthful feedback; both yield the same raw F-vs-N ordering. No pre-specified outcome was dropped, and all other reported measures — including the evolved trait change — are exploratory.

Table 3. Main paired comparisons (Wilcoxon signed-rank on per-seed differences; matched-pairs rank-biserial r ; bootstrap 95% CI of the median difference). FL = flagship (50 pairs), HS = harsh shock (30 pairs). “adj.” compares $\Delta P = P_{\text{actual}} - P_{\text{passive}}$ between conditions (Sect. 5.2).

Outcome	Comparison	Δmedian [CI]	r	p
all state outcomes (FL, HS)	B vs A	all differences $\equiv 0$	—	—
story pull, raw (HS)	F vs N	+0.0066 [0.0037, 0.0095]	0.89	1.7×10^{-6}
story pull, adj. (HS)	F vs N	+0.0022 [−0.0003, 0.0034]	0.36	0.084
story pull, adj. (FL)	F vs N	+0.0004 [−0.0016, 0.0022]	0.08	0.63
cooperation (FL)	C vs A	+0.200 [0.172, 0.239]	1.00	8.9×10^{-15}
cooperation (HS)	C vs A	+0.209 [0.169, 0.248]	0.99	9.3×10^{-9}
fragmentation (FL)	F vs A	−0.017 [−0.034, −0.006]	−0.52	9.5×10^{-4}
$\Delta\bar{\gamma}$ (HS)	F vs A	−0.336 [−0.456, −0.281]	−1.00	1.9×10^{-9}
$\Delta\bar{\gamma}$ (HS)	N vs A	−0.323 [−0.431, −0.240]	−0.94	2.0×10^{-7}
recovery time (FL, HS)	C, F, N vs A	$\Delta\text{median} = 0$ $ r < 0.5$		> 0.22

Passive counterfactual. Crucially, P can also be evaluated on trajectories that never received the reference: for each treated run we compute P on the paired-seed *untreated* trajectory (A/B) scored against a matched reference of the same type, defining the *counterfactual performativity gap* $\Delta P = P_{\text{actual}} - P_{\text{passive}}$ (Sect. 5.2). Campaigns: a *flagship* run (conditions A/B/C/ F_{invert} /N $\times$ 50 paired seeds, mild shock), a *harsh-shock* replication (40% hub removal, $\times$ 30 seeds), a *sensitivity sweep* (feedback gain $g \in \{0.2, 0.8, 1.6\} \times \{A, C, F_{\text{invert}}, F_{\text{crisis}}, F_{\text{utopia}}, N\} \times$ 20 seeds), a $2\times$ *scale check* (doubled population and resources), a conformist *response-regime campaign* (15 seeds per cell), and a *replay control* (30 receiving runs on the harsh-shock seed list plus 15 auxiliary source runs): the core campaigns comprise $250 + 150 + 360 + 90 + 32 + 30 + 15 = 927$ runs of 3,000–4,000 steps; the two architectural controls (Sect. 5.5) add 300 each, for 1,527 in total. Because conditions share seed lists the design is paired; the primary analysis is a Wilcoxon signed-rank test on per-seed differences with matched-pairs rank-biserial r and a bootstrap 95% CI on the median difference. Unpaired Mann–Whitney U with Cliff’s δ is a secondary robustness view; all inferential claims follow the paired analysis, and other logged measures are exploratory.

5 Results

Table 3 summarizes the main paired comparisons; the following subsections unpack them.

5.1 Measurement alone is causally inert

Under paired seeds, condition B is trajectory-identical to A: every per-seed difference is exactly zero, on every outcome, in both campaigns. The instrument acquires causal power only when it broadcasts.

5.2 Apparent story pull is explained by a passive counterfactual

Except where stated, results concern the corrective regime; Sect. 5.4 shows how they change under conformity.

Evaluated naively, the story-pull statistic appears to show strong self-fulfilling steering. Raw P (harsh shock; medians, bootstrap 95% CIs) is positive in all three non-truth signal conditions — F 0.0078 [0.0066, 0.0100], N 0.0014 [0.0001, 0.0023] — and largest for the replayed genuine self-model of a sibling world (condition R; 30 receivers on the harsh-shock seed list, one of 15 source trajectories each): 0.0241 [0.0197, 0.0265]. The raw hierarchy $R > F > N$ is highly reliable (F vs N $r = 0.89$, $p \approx 1.7 \times 10^{-6}$; robust to one receiver per source, $n = 15$, $r \geq 0.98$, and to a source-clustered bootstrap).

A short observation shows why raw P can mislead. Under the inverted reference $b_t = 1 - s_t$, $\text{sign}(b_t - s_t) = \text{sign}(\frac{1}{2} - s_t)$; if an untreated component simply mean-reverts toward mid-range, $s_{t+1} - s_t = -\kappa(s_t - \frac{1}{2}) + \eta_t$ with $\kappa > 0$ and $\mathbb{E}[\eta_t | s_t] = 0$, then $\mathbb{E}[\text{sign}(b_t - s_t)(s_{t+1} - s_t) | s_t] = \kappa |s_t - \frac{1}{2}| > 0$: ordinary mean reversion masquerades as attraction toward the mirror, with zero causal feedback; post-shock relaxation plays the same role for the replayed reference, which tracks where recovering trajectories go anyway. The remedy is the *counterfactual performativity gap* ΔP : score the paired-seed *untreated* trajectory (A, or the identical B) post hoc against a matched reference of the same type. Untreated worlds “move toward” these references *more* than treated worlds move toward their actual broadcasts — passive medians 0.0119, 0.0300, 0.0073 — giving uniformly negative ΔP : -0.0036 (F), -0.0058 (R), -0.0055 (N), each $p < 2 \times 10^{-5}$. Between-condition contrasts of ΔP are not significant in either campaign (F vs N: harsh $p = 0.084$; flagship $p = 0.63$; R vs N: $p = 0.95$). Raw P is thus dominated by the intrinsic dynamics above, which align best with realistic references and worst with noise (excluding the shock transition shifts every median by < 0.0004 absolute and changes no conclusion). Response activation confirms the reading: F keeps four of five corrective rules active on $\geq 90\%$ of ticks while R is mostly quiescent ($\langle |b_k - s_k| \rangle$: 0.64, 0.36, 0.10 for F, N, R), yet R shows the largest raw pull — which activation therefore cannot explain. Under the primary corrective regime Eq. 2 thus provides *no evidence of aggregate content-specific attraction beyond intrinsic dynamics*; the small negative ΔP , similar across signal types, is consistent with broadcasts mildly dampening that drift. This is the paper’s methodological result: a statistic natural in empirical performativity studies registers large, internally consistent, and entirely artifactual “steering” unless checked against passive twins.

Structural spillovers, by contrast, are real. Direct outcome contrasts are unambiguous: agents persistently told the network is fragmenting over-connect it (mean degree 19.8 vs 12.3 in matched flagship runs at $t = 2000$; paired $\Delta \text{median} +7.2$, $p \approx 7 \times 10^{-10}$), reducing post-shock fragmentation (flagship paired $\Delta \text{median} -0.017$, $r = -0.52$, $p \approx 10^{-3}$; directionally consistent, ns, under harsh shock). Broadcasts densify and flatten the network: relative to A (paired, harsh shock, per-run medians over $t > 500$), mean degree rises $+1.1$ under truth

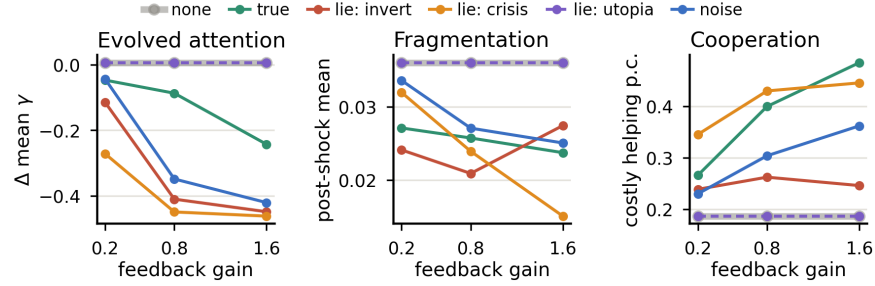


Fig. 2. Sensitivity sweep (40% hub-removal shock; 20 seeds per cell; medians). Left: evolved change in mean γ — attention varies strongly with broadcast type and gain, and the flattering (utopia) lie tracks the no-feedback baseline exactly (dashed over grey). Middle: under corrective responses the alarmist (crisis) lie does *not* come true. Right: deficit signals mobilize costly helping.

($r = 0.49$), +3.8 under the inverted lie ($r = 0.80$), +2.8 under noise ($r = 0.73$), while Freeman centralization (F: -0.026 , $r = -0.74$; N: -0.010) and betweenness concentration (F: -0.007 , $r = -0.67$; N: -0.008) fall: the fed-back network is denser and less hub-dominated than the silent one.

5.3 The type of lie sets what changes

The three distortion modes produce qualitatively different *outcome* regimes (Fig. 2). The *inverted* self-model, always reporting the mirror of reality, keeps most corrective responses persistently active (four of five on $\geq 90\%$ of ticks): it mobilizes costly helping (+0.106 over baseline, flagship $r = 0.82$) and drives the over-connection above. The *crisis* self-model is self-defeating for structure — corrections push fragmentation away from the alarmist report — yet its permanent cooperation-deficit signal mobilizes the highest costly-helping rates of any lie under correction (0.35–0.45 across gains; only truthful feedback at $g = 1.6$ exceeds them, 0.48). The *utopia* self-model is behaviorally inert: because responses fire only on reported deficits, a flattering broadcast triggers nothing and is indistinguishable from disconnecting the channel, at every gain: under purely corrective rules, flattery equals silence. Caveat: these first-order response *directions* follow from the rules (Eq. 1) by construction; not built in are the magnitude asymmetry across distortion types, the structural spillovers, and the evolutionary consequences below.

5.4 The response regime decides which lies come true

Conformist agents reverse the taxonomy exactly as the mechanism predicts (Table 4). The *crisis* lie, self-defeating under correction, becomes strongly self-fulfilling under conformity: told the world is fragmenting, imitators cut ties and fragmentation rises an order of magnitude above baseline (0.400 vs 0.038). The

Table 4. The same lie under two response regimes (medians over the 15 seeds shared by both campaigns; harsh shock, $g = 0.8$). Baseline (A), identical in both regimes: cooperation 0.183, fragmentation 0.038.

Lie	Corrective		Conformist	
	coop.	fragm.	coop.	fragm.
Invert	0.256	0.020	0.419	0.054
Crisis	0.414	0.018	0.255	0.400
Utopia	0.183 (inert)	0.038 (inert)	0.428	0.025

utopia lie flips from behaviorally inert to benevolently self-fulfilling: “everyone cooperates” becomes true by imitation (cooperation 0.428). Under conformity even noise mobilizes: imitating randomly reported cooperation levels raises costly helping to 0.445, as high as any lie, since imitation amplifies whatever is reported. Which false self-models come true — in the outcome sense — is therefore a property of the lie–response pair, not of the lie alone. Compactly: with macrostate update $S_{t+1} = \Phi(S_t, R(B(S_t)))$ for broadcast map B and response rule R , self-fulfillment is a property of the composition $R \circ B$, not of B alone.

5.5 Attention declines under consequential broadcasts

Mean heritable global sensitivity $\bar{\gamma}$ changes over ~ 20 generations of lineage depth as an emergent, unprogrammed function of broadcast quality (Fig. 3). At $g = 0.8$ (harsh shock): no broadcast -0.03 (neutral drift), truth -0.10 , noise -0.34 ($r = -0.94$, $p \approx 2 \times 10^{-7}$), false -0.42 (Δ median -0.336 $[-0.456, -0.281]$, $r = -1.00$, $p \approx 1.9 \times 10^{-9}$). The sweep is dose-ordered: the decline under the inverted broadcast deepens from -0.12 ($g = 0.2$) to -0.45 ($g = 1.6$) — and at the highest gain even *accurate* feedback shows a marked decline (-0.24). The categorical ordering compresses into one continuous relationship: with broadcast-implied corrective-drive intensity $I = g \langle \sum_a f_a(b(t)) \rangle$ (computable from the broadcast alone), evolved attention tracks I across the 15 sweep cells almost perfectly (Spearman $\rho = -0.99$; also strong at run level, $\rho = -0.83$, $n = 300$). Utopia ($I = 0$) is untouched; truth exhibits less decline where its corrective drive is smaller at equilibrium; unreliable high-drive broadcasts pay most. Corrective-drive intensity, rather than veracity alone, appears to organize the response. Since I omits the per-agent factor γ_i (the quantity under selection) and is computed from realized rather than independently randomized trajectories, we treat it as descriptive, not as causal mediation.

Not an architectural artifact. Two asymmetries in the response architecture could in principle manufacture this pattern. First, feedback adds weight only to non-reproductive actions, so $P(\text{select reproduce} | \gamma_i, b) = w_r / (W_0 + \gamma_i g D(b))$ falls with γ_i whenever $D(b) > 0$ — and I governs exactly that penalty. Second, hub-targeted pruning fires with probability γ_i rather than $\gamma_i g$, giving sensitivity a second, differently scaled channel. We reran all five broadcast conditions against

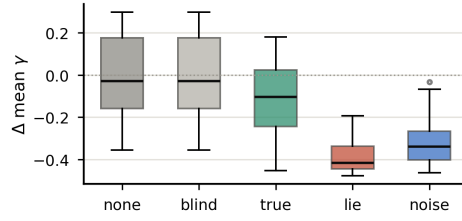


Fig. 3. Evolution of attention to the self-model (harsh shock, $g = 0.8$, 30 seeds per condition): change in mean $\bar{\gamma}$, first to final observer record. *none* and *blind* are identical by construction. The ordering (*lie* < *noise* < *true* < *none*) emerged from the evolutionary dynamics alone; nothing in the rules references broadcast reliability.

each (300 runs apiece): a reproduction-neutral variant rescaling the reproduce weight by $(1 + \gamma_i g D / N_0)$, holding that probability exactly at its no-broadcast value while leaving other feedback weights unchanged; and a variant fixing hub-pruning probability at 0.5 for every agent. Neither changes the result: median retention across all cells with a negative baseline $\Delta \bar{\gamma}$ is 98% and 100% respectively, the intensity correlation persists ($\rho = -0.98, -1.00$), and at $g \geq 0.8$ truth in fact declines *more* under the reproduction control. Selection on γ is ecological rather than an artifact of either channel; the remaining pathway is not explained by a negative marginal fecundity gradient (within-run $\rho(\gamma_i, \text{offspring}) = +0.09$ under F, against -0.05 with no broadcast) nor by shock mortality (disabling the shock entirely leaves the decline at -0.40 against -0.39 with it; both from a 5-seed side experiment), and is left open.

5.6 Truthful feedback doubles cooperation; recovery time is untouched

True feedback roughly doubles costly helping (flagship: 0.397 vs 0.179; paired $\Delta \text{median} +0.200$ [0.172, 0.239], $r = 1.00$, $p \approx 9 \times 10^{-15}$; replicated under harsh shock, $r = 0.99$), rising monotonically with gain ($0.27 \rightarrow 0.48$). The loop implements distributed corrective feedback: deficits reported at the population level are repaired by uncoordinated individual action. By contrast — a pre-specified null — recovery time showed no feedback effect in either main campaign, in any condition, even with 40% of hubs removed (medians 10 and 40 steps; all paired $\Delta \text{median} = 0$, $p > 0.22$). One lead runs the other way: at $2\times$ scale the inverted lie delayed recovery in every non-tied pair (+10 steps, $r = 1.00$), though with $n = 8$ that is the smallest attainable p (0.0625) and it does not replicate at base scale. Resilience is thus predominantly demographic rather than informational.

5.7 Robustness to system scale

Re-running the core conditions (A, C, F_{invert} , N) at doubled population and resources (steady-state $\approx 120\text{--}130$ vs ≈ 60 ; $n = 8$ pairs) preserves the main

effects: cooperation under truth 0.391 vs 0.173 (Δ median +0.151 [0.119, 0.247], $r = 1.00$, $p = 0.008$) and the attention ordering (F vs A -0.299 [$-0.388, -0.212$], $r = -1.00$, $p = 0.008$; N vs A -0.219 , $r = -0.89$, $p = 0.023$).

6 Discussion and Limitations

As noted alongside the results, the first-order response directions are encoded in the rules; the findings live one level up. First, the methodological lesson: a statistic that looked like compelling evidence of self-fulfilling dynamics — positive, replicated, and largest for realistic signals — was fully explained by intrinsic dynamics on passive twin trajectories. Empirical tests of performativity face this identification problem [22], typically without access to the untreated twin our platform provides. Second, the substantive findings: broadcasts change behavior and structure with a regime-dependent taxonomy of lies, and above all the evolutionary layer, in which the credibility of a collective self-model becomes an evolving property of the population. The response mappings are controlled mechanism probes, not calibrated behavioral models: they isolate how alternative receiver transformations turn identical broadcasts into different outcomes; whether learned or empirically inferred policies preserve the taxonomy is open. Further limitations: a single ecology and parameter family (though effects persist across an $8\times$ gain range and a $2\times$ scale change), and one network per run rather than interacting populations.

The results have a practical reading. Metric-driven infrastructure (autoscalers, observability pipelines, trending algorithms) is exactly this loop: the false-broadcast conditions model telemetry corruption, and under deficit-driven responses *flattering telemetry can be operationally equivalent to no telemetry* while raising no alarms. Reactive-measurement episodes such as institutional ranking reactivity [4] are natural empirical targets — with the passive-counterfactual check as the entry requirement. For multi-agent AI, where language-model agent populations [15] make shared collective-state dashboards natural, the evolution-of-distrust result suggests persistent misreporting does not merely misdirect a population — it degrades the control channel itself.

7 Conclusion

We introduced a reproducible causal laboratory for reflexive self-modeling in adaptive networks and showed that (i) the measurement-only control is trajectory-identical to baseline, while broadcasting changes behavior, structure, and evolution; (ii) apparent self-fulfilling “story pull” — strong, replicated, largest for realistic signals — vanishes against a passive counterfactual, a caution for empirical performativity claims; (iii) which lies come true, in the outcome sense, is set by the lie-response pair, not the lie alone; and (iv) heritable attention to the collective self-model declines across generations, tracking the broadcast-implied corrective-drive intensity — a population’s trust in its own self-model is

an evolving quantity that high-drive misleading feedback spends down. Future work: learned response policies and language-model agents.

Reproducibility. Code, configurations, analysis scripts, pinned dependencies, and deterministic regeneration of every run: github.com/Ivan825/Project-ONE, tag `cn2026-submission`; stored run records and replay trajectories ship as release assets there.

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